19 A hot air balloon is descending.
The height of the balloon n minutes after it starts to descend is h_n metres.

The height of the balloon $(n+1)$ minutes after it starts to descend, h_{n+1} metres, is given by

$$h_{n+1} = K \times h_n + 20 \quad \text{where } K \text{ is a constant.}$$

The balloon starts to descend from a height of 1200 metres at 09 15
At 09 16 the height of the balloon is 1040 metres.

Work out the height of the balloon at 09 18

09:16: 1040 = 1200K + 20.

K = (1040 – 20)/1200 = 0.85.

09:17: h = 0.85 × 1040 + 20 = 904 m.

09:18: h = 0.85 × 904 + 20 = 788.4 m.

4 marks total
Substitute first two heights
K=0.85
Two recursive steps
788.4 m

..... m

(Total for Question 19 is 4 marks)

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